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The predictive validity of selection and training measures on training performance of Air Traffic Controller trainees

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Abstract

The job of an air traffic controller (ATCO) is very specific and demanding. The assessment of potential suitable candidates requires a customized and efficient selection procedure. The German Aerospace Center DLR conducts a highly-selective, multi-stage selection procedure for ab initio ATCO applicants for the German Air Navigation Service Provider DFS. Successful applicants go on to complete an initial training at the Academy (ca. 13 to 17 months) and then a unit training phase in live traffic (ca. 18 months). ATCO validity studies are scarcely reported in the international scientific literature and have mainly been conducted in a military context with only small and male samples. The validation study at hand encompasses the complete data from 476 DFS ATCO trainees starting with candidate selection and extending to the completion of their training. Univariate and multivariate data analyses involved the prediction of training success (i.e., pass/fail) and training performance criteria, derived from initial training. The final training success rate of about 80% was rather high compared to other countries. The findings demonstrate that cognitive ability measures predicted performance in initial training and exam repetitions. Contrary to prior studies, Assessment Center (AC) exercises did not meaningfully contribute to the prediction of training success but were helpful for generating hypotheses for the semi-standardized interview. The interview provided additional validity over and above cognitive ability tests and work-sample tests. Performance in the initial training was positively related to successive training performance.

Introduction

Due to the low base rate of suitable applicants, the high importance of safety for the job and the high costs of ATC training the selection of ab initio air traffic controller (ATCO) trainees requires a valid and efficient approach to recruitment.
The selection procedure for ATCO trainees for DFS German Air Navigation Services has been developed and conducted in close cooperation with DLR German Aerospace Center. The selection process involves multiple stages, starting with a paper sift based on published application criteria (e.g. age, education, language training). The first stage of testing covers a comprehensive set of mental abilities (i.e., memory, attention, processing speed, English), followed by a stage of work-sample testing (two multiple-task performance tests). The third stage focuses on teamwork abilities (two exercises). The final phase involves an oral English test and a semi-structured biographical interview. Successful trainees undergo a medical examination according to Eurocontrol class 3 requirements. The entire selection process follows an empirically based selection strategy in order to ensure a sufficient aptitude level for all aspects of the aptitude profile. For each stage and the final selection decision (i.e. recommended for ATCO training at DFS or not recommended), clearly defined cut-off scores and compensation mechanisms are applied. The testing procedure results in an empirical selection rate of about 6%.
The continuous and intense quality assurance of the process is based on four major pillars: (I) job requirement analyses (e.g., Bruder, Jörn, & Eißfeldt, 2008; Eißfeldt & Heintz, 2002), (II) proper test system development, careful selection and regular training of the psychological and operational staff involved (Seidel, Pecena & Eschen-Léguéde, 2009), (III) frequent and detailed cost-benefit analyses (e.g. Heintz, 2004) and (IV) regular psychometric validation. The validity of this DLR selection program has already been proved in two former large scale validation studies (Damitz, Eißfeldt, Grasshoff, Lorenz, Pecena & Schwert 2000; Eißfeldt & Maschke, 1991). These previous studies resulted in adaptations and further developments of the respective test program. In addition, several validity analyses have been conducted on individual tests or methods. For example the reliability and validity of the team work related Assessment Center approach (cf. Pecena 2003) has been carefully analyzed since its introduction in 1996, revealing high and significant correlations with training performance and

training success. Other intermediate validation analyses focused on the predictive validity of work-sample tests (e.g. Grasshoff, 2006), the predictive power of biographical factors and the long-term prediction of stress resistance in Air Traffic Controllers (Eißfeldt, Weber, Udovic & Vogt, 2006). In international scientific literature, several other validation studies in the field of air traffic controller selection have been reported (for an overview see Broach & Manning, 1998). However, the samples are usually small and there are hardly any which provide a validation of the full process, i.e. from the initial selection all the way to the completion of on-the-job training.
Therefore, the current study primarily aims at analyzing the complete selection procedure applied from 1997 to 2006, including the reliability and validity of the training process as a whole. Given this long period of time, a massive data set emerged from a diverse range of variables relevant to the selection and training process. Heintz et al. (2010) describe the design of this validation study in detail as well as the challenges and efforts to set up the appropriate data base. The initial findings have been reported by Conzelmann et al. (2011). In the paper at hand the main predictive findings are described.

Methods

The total validation sample consists of 476 ATCO trainees (mean age 20.52, S=1.74; 66%male) who were selected between 1997 and 2006. The trainees in the validation sample were trained using two subsequent stages of DFS training system DATS 1 (N=430) and DATS 2 (N=46) (DATS, DFS Air Traffic Controller Training System) and for different combinations of working position licenses (Aerodrome N=91 vs. En route and Approach N=385). Therefore, in addition to the total sample, several subsamples emerged, differentiated by training system and licenses. However, the sample size of DATS 2 was too small and only one trainee failed thus precluding any sophisticated validity analyses for this subsample. Therefore, the analyses will focus on the DATS 1 sample. The findings with this sample were cross-checked by the subsample of En route controllers. Furthermore, sample sizes of the analyses vary slightly. These variations are caused by missing data; first, on the predictor side, because the selection procedure was adapted during the course of the data collection and second, on the criteria side, because trainees dropped out at varying points during the training.
Different predictor variables were utilized for the validation analyses, depending on the different selection stages. For the pre-selection stage, composite scores were applied for each ability domain (memory, attention, processing speed). Performance on English examinations, work-sample tests, and Assessment Center exercises served as predictors. During the final semi-structured interview, the selection board evaluates applicants suitability for training based on aspects of their general motivation, their vocational motivation, and their social skills. Mean ratings of these domains served as predictor variables for the interview selection stage.
There was a vast number of diverse training variables from the initial training (IT) serving as criteria, whereas only a few criterion variables were available from unit training. Aside from the number of available variables, IT criteria are comparable among the trainees: all trainees attend the same courses during IT at the DFS Academy opposed to the individualized training for different sectors and units which they receive during UT. As the most important criterion, training success was differentiated between failure during IT, failure during UT, and overall training success that does not consider the point of failure in training. For IT, two composite scores were computed one summarizing performance in all theoretical examinations and the other summarizing performance in all practical (simulation) examinations. Whether or not an exam had to be retaken was also utilized as a criterion variable for the validation analyses. Additionally, in IT simulation exams, trainees were rated by the instructors on the 12 DATS evaluation criteria. Based on these ratings, a composite score was built by averaging the different criteria into one composite DATS IT instructor rating.
The predictor data used for the analyses was collected over a course of nine years, hence, different versions of selection tests were administered (i.e., variations in item material, number of items, and test duration). In general, raw scores from different test versions lack comparability. Therefore Stanine scores were utilized instead of raw scores for the major analyses. Due to the nature of the selection process where applicants with insufficient performance scores are systematically rejected, the predictor data lacked a normal distribution. Furthermore, criterion data displayed a skewed distribution. The full range of scales was rarely exhausted on the side of criterion variables, i.e. trainees generally received positive performance ratings, leaving the data distribution both restricted in range and skewed. Therefore, non-parametric statistics were applied including methods such as the Mann-Whitney U-Test, Spearman correlation, and the more complex logistic regression analysis. It should be noted that this rather conservative approach, as well as the range restriction in both predictor and criterion data, leads to a lower probability of finding significant results and also caps the statistical effect level of significant findings that can be achieved. Because of the different methods applied, the results reported here will focus on the significance level of the findings. Table 1 provides an overview of predictor and criterion variables.

Table 1. Summary of predictor and criterion variables

Predictor Variables	Criterion Variables	
Test Stages	**Composite scores**	
Biographical Information: Age, Gender, Education, Grade Point Average etc. • Pre-Selection (Memory, Attention, Processing Speed) • Work-Sample Tests • AC Team Exercises • English Examination • Semi-Structured Interview (General Motivation, Vocational Motivation, Social Competencies)	Initial Training (IT)	• IT Theory Score • IT Simulation Score • DATS IT Instructor Rating • Repetition of Examinations • Success in IT
	Unit Training (UT)	• Success in UT
	Overall success	• Overall success in training

Results and Discussion

Overall, 81% (DATS 1: 79%) of the trainees in the validation sample successfully completed their training as air traffic controllers. The success rate of aerodrome controller trainees (89%) was higher than that of en route and approach controller trainees (79%). Comparing UT and IT, the success rate was higher for IT (93%) than for UT (87%). 1% of the trainees had to discontinue training after failing a medical test; these instances were also counted as failure.
A significantly higher success rate was observed for female trainees, (89 % compared to 76 % for male trainees). Regarding age differences between the trainees, whereas the success rate for 18-19 year-old (87 % of N=243) and 20 year-old old trainees (86 % of N=157) exceeded the overall success rate, only 61 % of the trainees with an age of 23 or older (N=59) succeeded training.

Predictive Validity of the Selection Procedure

In personnel selection, predictive validity is the ultimate benchmark: 'How well does the test battery predict a candidates' performance during future training and on the job?' As described above, an extraordinarily comprehensive data set was available for this study. In the following, an overview of the main findings is provided (see Table 2). Asterisks in the table refer to repeated significant findings from various statistical analyses (group comparisons, correlations, linear and logistic regression). Findings that surpassed the .05 significance level ($p < .05$) are marked with a single asterisk and findings that surpassed the .01 significance level ($p < .01$) are marked with two asterisks.
Concerning the ability domains tested in the pre-selection stage, attention and processing speed tests contributed the most and to the greatest extent. The memory ability domain tests were relevant for predicting performance in more scholastic contexts, such as the IT examinations, which are reflected in the IT Theory score. To summarize, pre-selection tests, which test for a broad spectrum of cognitive demands, proved useful for almost all criteria variables. These results support the notion that trainees have to possess a certain level of cognitive abilities, which are tested by the first pre-selection stage, and that these abilities, qualitatively as well as quantitatively, are necessary in order to complete the ATCO training successfully.
As shown in Table 2 significant results that indicated predictive validity of work-sample tests were obtained for simulation performance in early stages of IT. Associations between work-sample test performance and the IT Theory score were only marginal and scattered. Regarding the IT Simulation Score and the DATS IT Instructor Rating, significant associations were found for the work-sample tests. Better performance in the work-sample tests corresponded with better performance ratings from

instructors during simulation training. DATS criteria such as Problem Recognition, Traffic Planning, Situation Awareness, Coordination Procedure, and Coping under Pressure were the main criteria that work-sample tests corresponded with. Given the face validity of the work-sample tests, the lack of predictive validity for UT may seem surprising at first. However, two aspects have to be considered: First, work-sample test are complex simulations of real future job demands. Their complexity is their strength; however, analyses based on a single final score might be misleading. Whether the validity of work-sample tests can be improved by adjusting the scoring procedure is a research question that will be addressed empirically. The results of sub-scores from this validation study support the notion that there is still certain information held within these sub-scores which is not yet completely reflected in the final test scores. Second, work-sample tests are administered during the selection procedure in order to anticipate demands that are encountered later in UT. Although there were various criteria from IT in this validation study, only few criteria from UT were available, and those lacked variance. A qualitative evaluation of trainee performance beyond just the successful completion of training is not yet available for the on-the-job training phase. Such criteria might be more suitable for validating work-sample tests. The significant findings between work-sample tests and DATS IT Instructor Ratings, which represent an evaluation of trainees' performance in IT simulation, support the idea that the work-sample tests have more predictive power for the operational on-the-job performance of trainees. Furthermore it has to be considered that selectivity of the selection stage dealing with work-samples is very high, leading to a particular high restriction of range.

Table 2. Overview of predictive validity of selection stages regarding different training criteria

Selection Stages		IT Theory Score	IT Simulation Score	DATS IT Instructor Rating	Repetition of Exams	IT Success	UT Success	Overall Success
Pre-Selection Stage	Memory	**						
	Attention	**	**	*	*		**	**
	Processing Speed	**			*	*	*	**
Work-Sample Tests			*	*				
AC Team Exercises			*	*				
English Examination		**						
Semi-Structured Interview	General Motivation	**	**	*	**	*	*	**
	Vocational Motivation	**			*	*		*
	Social Competencies		*	*			**	**

Note. Sample N=330-430, only repeated significant findings from different statistical analyses (group comparisons, linear and logistic regression) are reported; *p<.05 and ** p<.01

Only few significant results were found for the scores from the Assessment Center team exercises. Small but significant associations were obtained for sub-scores from the team exercises with both the IT Simulation Score as well as the DATS IT Instructor Score. However, the low number of predictive results for the AC exercises contrasts with the findings from previous validity studies (Pecena, 2003). In this study AC performance has shown consistent predictive validity for training criteria such as training success as well as performance on theoretical and practical examinations. However, when compared to the current study, there are several core differences: First, in the previous study the AC team exercises were not treated as an independent selection stage, but instead were used to provide additional information on applicant behaviour. Only applicants with both extremely low AC performance ratings and questionable performance on work-sample tests were excluded from further testing. Consequently, the range of AC performance in the current validation sample is more restricted than in the previous validation study, thereby also preventing further significant findings. Second, DATS, the new DFS ATCO training system had not been implemented during the data collection

phase of the previous study. Therefore, different criteria data were available than in the current validation study. AC team exercises were not only used as a direct means of selection, but the resulting AC performance profile is also utilized for the semi-structured interview. Hypotheses are generated from the behavioural ratings and further explored in the final interview. This procedure is supported by analyses which confirm that AC performance has predictive value for the success in the following interview. Since the interview in its current form showed far-reaching predictive validity for many criterion variables, the AC performance indirectly aids this predictive capacity by providing preparatory information.

Performance in written and oral English examinations proved most predictive for IT Theory performance. As expected, the highest predictive validity was found for English Language Training during IT. In later stages of IT which involve more simulation training, English proficiency had no additional significant predictive validity for IT performance. These findings could be explained by differing English proficiency levels at the beginning of IT. Since some courses are conducted completely in English, trainees with higher levels of English proficiency would have less difficulty following the subject matter than trainees with lower levels of English language proficiency, who would have to put additional effort into understanding the language and following the subject matter. Due to additional English training during IT, language proficiency differences become marginal and less important to performance differences.

Results confirming the predictive validity of the interview ratings were found for all criterion variables, as shown in Table 2. General Motivation, Vocational Motivation and Social Competencies) proved useful in predicting performance in IT Theory, IT Simulation, DATS IT Instructor Rating and later training success. Differentiating further, Social Competencies were more valid for UT and the practical aspects of IT, whereas General Motivation and Vocational Motivation showed predictive validity for almost all aspects of IT performance. These findings highlight the importance of social and motivational aspects and should be considered thoroughly during the decision-making process. To summarize: the current procedure, in which the interview serves as a final selection stage, utilizing all information available in order to reach a final decision on the applicants' ability to begin ATCO training, is confirmed.

Predictive Validity of Initial Training Performance

For the analyses of the predictive validity of IT training performance, success in UT was used as the relevant criterion variable. Using overall training success would have also included trainees who had already failed training in IT, mostly due to poor performance. Since the question in focus is the extent to which UT success depends on IT performance, associations would be overestimated, if trainees who had already failed IT were included.

Significant associations were found between success in UT and both composite scores for IT. The IT Simulation score correlated higher with UT success (r_S=.18, p<.01) than the IT Theory score did (r_S=.12, p<.01). For both scores, better performance in IT corresponded to a higher probability of success later in UT. The relation between UT success and the averaged DATS IT Instructor Rating was also significant (r_S=-.13, p<.01). Again, better ratings from instructors were related to a higher probability of success in UT.

Success in UT also showed a significant correlation with repetition of exams (r_S=-.11, p<.05). Trainees who had to repeat at least one exam during their IT were more likely to fail in UT. Table 3 provides a descriptive overview of the UT success rate and exam repetition. 48.7% of trainees repeated at least one exam. Failure rates were higher, but still on a satisfactory level, for trainees who had repeated an exam during IT (18.6%) than for those who had not (10.8%).

Table 3. Success rate in UT related to repetition of exams

	Success in UT		
	Fail	**Pass**	**Total**
No Exam Repeated	22 (10.8%)	182 (89.2%)	204 (51.3%)
Repetition of Exam(s)	36 (18.6%)	158 (81.4%)	194 (48.7%)
Total	58 (14.6%)	340 (85.4%)	398

Note. Sample DATS 1.

To summarize, trainees were more likely to succeed in UT if their IT performance was better and if they did not have to repeat exams in IT. The association between IT performance and UT success

was higher for Simulation performance and the DATS IT Instructor Rating than for theoretical examinations. Since UT mainly consists of operational on-the-job training, the stronger association with simulation performance than with the IT Theory score is not surprising and thus also supports the importance and validity of simulation training preceding UT.

Conclusion

This validation study was ambitious in several aspects. The combination of sample size, number of variables involved, the long validation horizon and the degree of detail in the analyses are unparalleled by previous validation studies in the field of ATC. Given the low likeliness for significant results in terms of classic, correlation-based validity analyses due to the high selectivity of the selection process and the low variance in training performance data, all stages of the selection process turned out to be predictive for the complete training and licensing process. The relevance of Initial Training for later success in Unit Training could also be confirmed. Additionally, further internal reports address the selection as a multi-stage process in which subsequent stages depend on the previous selection stages. This step-by-step structure is one of the major strengths of the DLR selection procedure which has been developed and refined over the course of 30 years of selection experience. It helps ensure that the applicants who are selected have a strong aptitude profile for all aspects of the ATCO job requirements. Furthermore, it establishes an efficient selection procedure in terms of the cost-benefit ratio (Heintz, 2004).

In conclusion, the current selection procedure has proved valid for the selection of ATCO applicants. The total trainee success rate of almost 81% supports this claim and is seen as the key criterion for evaluation. Even if tests accurately predict training performance, the overall training success remains the foundation for evaluating the stages of recruitment and selection, as well as training and licensing. Only a high trainee success rate can lead to more cost effective training processes and enables reliable staff planning for the ATCO workforce.

The observed success rate has increased from the former validity study of the DLR selection procedure (67%, Damitz et al. 2000), even though traffic amount and complexity in Central European Airspace have increased over time. This success rate is also one of the highest achieved so far in comparable international reports (EATCHIP, 1996). Consistent and regular monitoring of validity is an important and indispensable principle for the success of selection and training procedures. Regularly monitoring the process, beginning with recruitment and throughout the training, not only helps ensure high success rates, but also provides valuable information which can be used to make improvements and adjustments to the process in the future.

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